MIL-F-36686
10 August 1970

MILITARY SPECIFICATION

FORCEPS, FIXATION, STRAIGHT, NONMAGNETIC, 4- 1/2 inch

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of nonmagnetic, fixation forceps, suitable for removing foreign bodies from the eye.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6515

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC) 10070 FORCEPS, SURGICAL, EYE

3. REQUIREMENTS

3.1 General Requirements. The fixation forceps shall be in accordance with the applicable requirements of GG-I-526, and as specified herein.

3.2 Material. Shall be fabricated from Class 3, Type 316 material.

3.3 Hardness. Rockwell hardness shall be B75 to B85.

3.4 Style, design and dimension. Shall be a nonmagnetic fixation forceps, having straight blades with meshing multiple teeth. Forceps shall be supplied with or without guide pin. Shall be suitable for use with FSN 6515-299-8008 MAGNET, EYE, ELECTRIC in removing foreign bodies from the human eye. Forceps shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 10070 for appropriate stock number and item identification.

3.5 Finish. Finish shall be mirror, or mirror and satin.

3.5.1 Passivation. The forceps shall be properly passivated.

3.6 Nonmagnetic characteristics. The forceps shall show no magnetic response when tested as specified in 4.4.1.

3.7 Identification marking. In addition to GG-I-526 requirements, foreign instruments shall be marked "stainless".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government: The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Shall be in accordance with GG-I-526.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the corrosion-resisting steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination purposes shall be one forceps.

Table I. Sampling Criteria for Examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. Unit of product for test purposes shall be one forceps.

Table II. Sampling Criteria for Tests

Component	Characteristic	Req.	Test procedure	Insp. level	AQL
Forceps	Composition analysis	3.2 and GG-I-526	GG-I-526	**	*
	Hardness	3.3 and GG-I-526	GG-I-526	S-2	*
	Dimensions	3.4	---	S-1	2.5
	Boil test	GG-I-526	GG-I-526	S-2	*
	Nonmagnetic test	3.6	4.4.1	S-2	*

* Acceptance number shall be zero for all sample sizes.

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** Four ounces of each lot of metal used in construction of the forceps shall be selected for chemical analysis.

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test requirements.

4.3 Examination. The forceps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Inspection, however, shall not be restricted to defects listed in Table III. Any forceps in the sample containing one or more defects shall be rejected and if the number of defective forceps in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Forceps not free from sharp edges (except for teeth), rough areas, burrs, cracks, tool marks and pits.
102	Finish not as specified.
103	Teeth not of uniform length and depth.
104	Teeth not well defined, or do not mesh properly when forceps are in closed position.
Minor	
201	Identification marking not present, correct, complete or legible.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Nonmagnetic test. The fixation forceps shall be placed at rest, laying flat, upon a smooth nonferrous surface. An Alnico horseshoe-shaped magnet, weighing between 0.131 and 0.145 pounds shall then be used to slowly approach, and make contact with, the surfaces of the forceps. The magnet shall be fabricated from Alnico No. 5 alloy, and shall be proven capable of supporting 43 cubic inches of steel (3-1/2 inches X 3-1/2 inches X 3-1/2 inches) weighing 12½ 1/2 pounds.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One forceps, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	288 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method 1 A-8.

5.1.3.5 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The fixation forceps are to be used in removing foreign bodies from the eye.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Applicable levels of packaging and packing (see 5.1 and 5.2)

(c) Federal stock number.

6.3 This specification is intended to cover Federal Stock Number 6515-333-9900 FORCEPS, FIXATION, Straight, Nonmagnetic, 4-1/2 inch, appearing in the Federal Supply Catalog.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity::
Defense Supply Agency - DM

6515-1491

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION: MIL-F-36686 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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MAR 68

EDITION AUG 66, WILL BE USED UNTIL EXHAUSTED

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